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問題1

図1のように，真空中（誘電率ε_0，透磁率μ_0）に完全導体からなる半径aおよびb（$a < b$）の無限に長い二つの導体円筒M，Nが同軸に配置されている．円筒の軸をz軸とする．円筒M，Nの厚みは無視できる．二つの導体円筒のz軸方向単位長さ当りの静電容量，自己インダクタンスを，それぞれCおよびLとする．以下の問に答えよ．

(1) MおよびNをz軸方向単位長さ当りそれぞれλ，$-\lambda$の電荷で一様に帯電させた．そのときに形成される電界$\boldsymbol{E}$を求めよ．また，静電容量Cを求めよ．

(2) MおよびNにz軸方向にそれぞれに一様な直流電流 $+I$および$-I$を流した．そのときに形成される磁束密度$\boldsymbol{B}$を求めよ．また，自己インダクタンスLを求めよ．

次に電流が時間変化する場合を考える．時刻t，位置zにおけるMとNの電位差を$V(z,t)$，MおよびNにz軸方向に流れる電流をそれぞれ $+I(z,t)$および$-I(z,t)$とする．

(3) $V(z,t)$のz軸方向の変化と$I(z,t)$の時間変化の関係を表す微分方程式を求めよ．C，Lを用いてよい．

(4) $I(z,t)$のz軸方向の変化と$V(z,t)$の時間変化の関係を表す微分方程式を求めよ．C，Lを用いてよい．

(5) 問(3)(4)の結果から，$V(z,t)$および$I(z,t)$がz軸方向に真空中の光速cで伝搬することを示せ．

(6) $I(z,t) = I_0\cos(\omega t - kz)$（但し$\omega > 0, k > 0$）と表せる時，$\omega$と$k$の関係を求めよ．

(7) 問(6)の電流が流れている時，円筒間の空間に形成される電磁場のエネルギー密度uとポインティングベクトル$\boldsymbol{S}$を求めよ．また，これらの結果からエネルギーuがz軸方向に真空中の光速cで運ばれていることを示せ．

必要であれば，円筒座標系(r,θ,z)における以下のベクトル解析の公式を用いて良い．

$$\mathrm{grad}\, f = \frac{\partial f}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\boldsymbol{e}_\theta + \frac{\partial f}{\partial z}\boldsymbol{e}_z$$

$$\mathrm{div}\, \boldsymbol{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

$$\mathrm{rot}\, \boldsymbol{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z}\right]\boldsymbol{e}_r + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\boldsymbol{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial \theta}\right]\boldsymbol{e}_z$$

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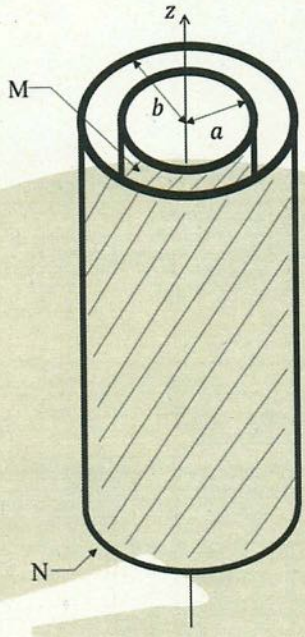


図 1

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Problem 1

As shown in Fig. 1, two infinitely-long perfectly-conducting cylinders M and N of radii a and b ($a < b$) are coaxially placed in a vacuum (permittivity ε_0, permeability μ_0). The axis of the cylinders is set as the z axis. The thickness of the cylinders is negligible. The capacitance and self-inductance per unit length along the z direction of the two conducting cylinders are defined as C and L, respectively. Answer the following questions.

(1) M and N are uniformly charged with the charge λ and $-\lambda$ per unit length along the z direction, respectively. Find the electric field $\boldsymbol{E}$ induced by the charge. Calculate the capacitance C.

(2) Direct currents $+I$ and $-I$ along the z direction are uniformly flowing in M and N, respectively. Find the magnetic flux density $\boldsymbol{B}$ induced by the current. Calculate the self-inductance L.

Next, consider the cases when currents flowing in the cylinders are changing in time. Define the voltage between M and N at time t and position z as $V(z,t)$ and the currents flowing in M and N along the z direction at time t and position z as $+I(z,t)$ and $-I(z,t)$, respectively.

(3) Derive the differential equation which expresses the relationship between the change of $V(z,t)$ along the z direction and the change of $I(z,t)$ in time. You can use C and L.

(4) Derive the differential equation which expresses the relationship between the change of $I(z,t)$ along the z direction and the change of $V(z,t)$ in time. You can use C and L.

(5) Using the answers to Questions (3) and (4), show that $V(z,t)$ and $I(z,t)$ are propagating along the z direction with the speed of light in vacuum, c.

(6) Suppose $I(z,t) = I_0 \cos(\omega t - kz)$ with $\omega > 0$, $k > 0$. Find the relationship between ω and k.

(7) When the currents in Question (6) are flowing, calculate the energy density u and the Poynting vector $\boldsymbol{S}$ of the electromagnetic field formed within the space between the cylinders. From these results, show that the energy u is carried along the z direction with the speed of light in vacuum, c.

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If needed, you may use the following vector analysis equations in the cylindrical coordinate system (r, θ, z).

$$\text{grad } f = \frac{\partial f}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\boldsymbol{e}_\theta + \frac{\partial f}{\partial z}\boldsymbol{e}_z$$

$$\text{div } \boldsymbol{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\theta}{\partial \theta} + \frac{\partial A_z}{\partial z}$$

$$\text{rot } \boldsymbol{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial \theta} - \frac{\partial A_\theta}{\partial z}\right]\boldsymbol{e}_r + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\boldsymbol{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial \theta}\right]\boldsymbol{e}_z$$

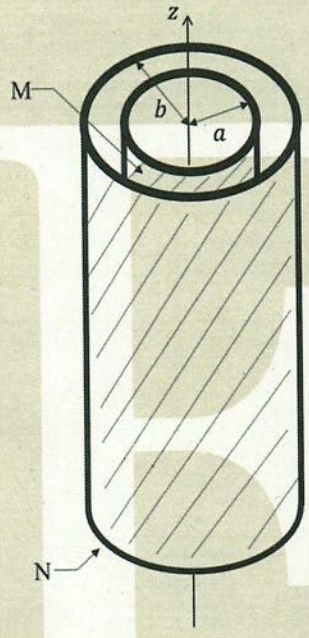


Fig. 1

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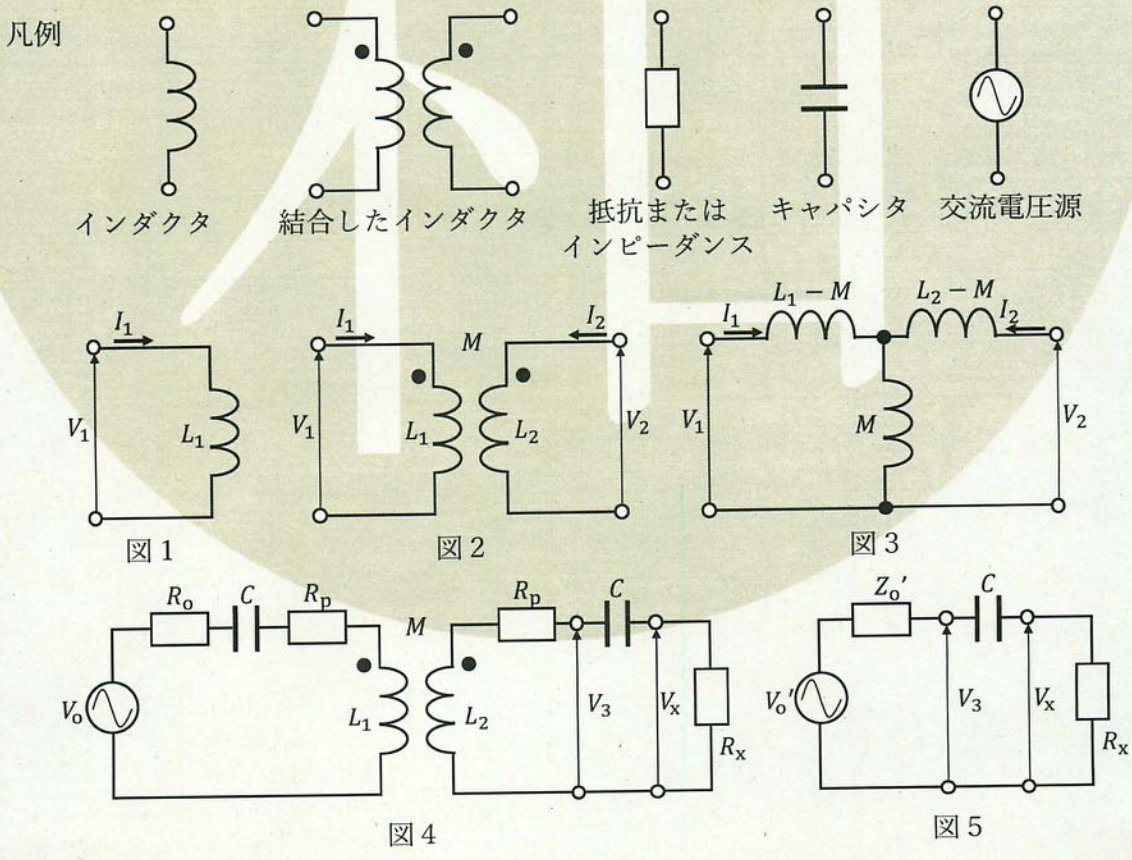
問題2

磁界共振結合による無線電力伝送回路について，以下の問に答えよ．図中の記号は凡例に従う．本問では，交流電圧，交流電流はすべて複素数表示とし，その絶対値は実効値を表す．

(1) 図1に示すインダクタL_1に，角周波数ωの正弦波交流電流I_1を印加した．インダクタ両端の電圧V_1を求めよ．
(2) 図2に示すように，結合したインダクタL_1，L_2に，角周波数ωの正弦波交流電流I_1，I_2をそれぞれ印加した．電圧V_1，V_2を求めよ．ただし，相互インダクタンスをMとする．
(3) 図2の回路と図3の回路とは交流に対して等価であることを証明せよ．

図4のように，電圧V_o，抵抗R_oの正弦波交流電源で無線電力伝送回路を駆動する．図中Cはキャパシタ，L_1，L_2は結合したインダクタ，R_p，R_xは抵抗である．以下の設問では$R_x = R_o = R$，$L_1 = L_2 = L$，および角周波数$\omega = 1/\sqrt{LC}$とする．

(4) 図4に示す回路と図5に示す回路とが等価である．V_o'，Z_o'を求めよ．ただし，Z_o'はインピーダンスである．
(5) 抵抗R_xの両端の電圧V_xとV_oとの比(すなわちV_x/V_o)を求めよ．
(6) V_x/V_oの絶対値が最大値を取るためのRの条件を求めよ．



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Problem 2

Answer the following questions on a wireless power transfer circuit using magnetic resonant coupling. The symbols in the figures can be found in the legend. In this problem, all sinusoidal AC voltages and currents are expressed as complex numbers, and their absolute values correspond to effective values.

(1) As shown in Fig. 1, a sinusoidal alternating current I_1 with an angular frequency ω is applied to the inductor L_1. Find the voltage V_1 across the inductor.

(2) As shown in Fig. 2, sinusoidal alternating currents I_1 and I_2 with an angular frequency ω are applied to the coupled inductors L_1 and L_2, respectively. Find the voltages V_1 and V_2. Note that their mutual inductance is denoted as M.

(3) Prove that the circuits in Fig. 2 and Fig. 3 are equivalent with respect to alternating currents.

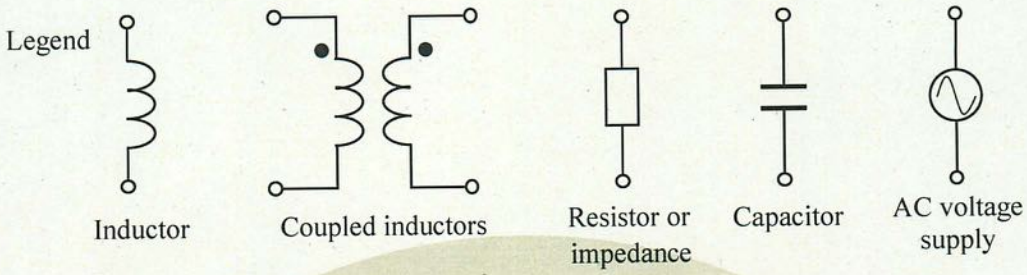
As shown in Fig. 4, a wireless power transfer circuit is driven by an AC power supply having a voltage V_o and a resistance R_o. In the figure, C is a capacitor, L_1 and L_2 are coupled inductors, and R_p and R_x are resistors. In the following questions, assume that $R_\mathrm{x} = R_\mathrm{o} = R$, $L_1 = L_2 = L$, and the angular frequency $\omega = 1/\sqrt{LC}$.

(4) The circuit in the Fig. 4 is equivalent to that in Fig. 5. Find V_o' and Z_o'. Note that Z_o' is an impedance.

(5) Find the ratio of the voltage V_x across the resistor R_x and V_o (i.e. $V_\mathrm{x}/V_\mathrm{o}$).

(6) Find the condition of R that maximizes the absolute value of $V_\mathrm{x}/V_\mathrm{o}$.

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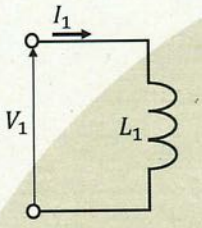


Fig. 1

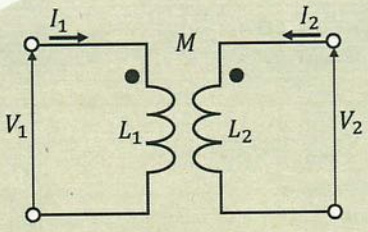


Fig. 2

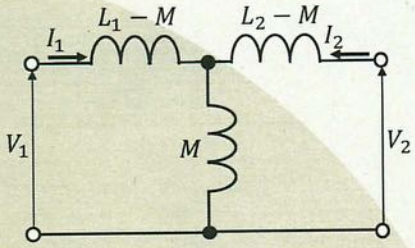


Fig. 3

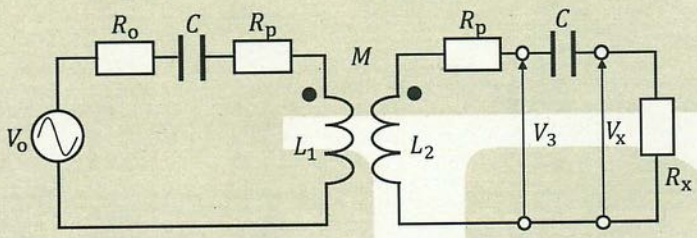


Fig. 4

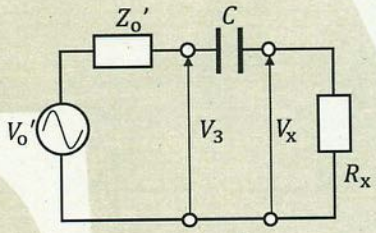


Fig. 5

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問題3

箱の中に赤と白のボールが入っている．一回の試行において，アリスは箱からボールを取り出し，色を確認してボールを戻す．アリスがキャロルを介してボブにどちらの色だったかを伝えることを考える．キャロルは気分屋であり，機嫌が良い時は 10%の確率で，機嫌が悪い時は 60%の確率で，嘘の色をボブに伝える．ただしアリス，キャロルともに発言する色は赤または白のみであるとする．キャロルは，機嫌が良い時に嘘をつくと機嫌が悪くなり，機嫌が悪い時に本当のことを言うと機嫌が良くなるものとする．キャロルの機嫌は良い，悪い，のいずれかである．この試行を複数回繰り返す．このとき以下の問に答えよ．必要に応じて $\log_2 3 = 1.585$，$\log_2 5 = 2.322$，$\log_2 7 = 2.807$を用いよ．

(1) 箱の中に赤いボールが 10 個，白いボールが 5 個入っている．箱を情報源とみなした時のエントロピーを求めよ．

(2) キャロルの機嫌が良い時を状態S_0，機嫌が悪い時を状態S_1とする．キャロルが嘘を言うと 1，本当のことを言うと 0 が出力されるような情報源を考えた時，この情報源の状態遷移図を示せ．

(3) 試行を十分な回数繰り返した後の定常状態において，キャロルの機嫌が悪い確率を求めよ．

(4) 問(2)の情報源について，そのエントロピーを求めよ．

(5) 次に，アリスがデイブを介してボブにどちらの色だったか伝えることを考える．デイブは問(3)で求めたものと同じ確率で嘘をついてボブに反対の色を伝えるとする．キャロルを介した試行の場合と比べて，ボブが受け取る情報量はどのように変化するかを示せ．またその理由を通信路容量という言葉を用いて説明せよ．

ここで，機嫌を正確に計測できるセンサをキャロルに取り付ける．複数回の試行ごとのキャロルの機嫌の組み合わせを，0, 1 からなる 2 元符号で符号化したい．

(6) 2 試行毎の機嫌の組み合わせに対して符号語を割り当てるとき，最も効率の良い符号を設計せよ．その時の 1 試行あたりの平均符号長を求めよ．

(7) 機嫌を組み合わせる試行回数を増やしたとき，この符号の 1 試行あたりの平均符号長の下限を求めよ．

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Problem 3

There are red and white balls in a box. In a trial, Alice picks up a ball from the box, checks the color of the ball, and returns it back to the box. Consider that Alice informs Bob of the color of the ball through Carol. Since Carol is fickle, she tells Bob the wrong color with a 10% probability when she is in a good mood, and with a 60% probability when she is in a bad mood. Note that the colors Alice and Carol utter are only red or white. If Carol tells Bob the wrong color when she is in a good mood, her mood becomes bad. Likewise, if she tells Bob the correct color when she is in a bad mood, her mood becomes good. Carol is only ever in a good or a bad mood. This trial is repeated multiple times. Answer the following questions. Use the following values if necessary: $\log_2 3 = 1.585$, $\log_2 5 = 2.322$, $\log_2 7 = 2.807$.

(1) There are ten red balls and five white balls in the box. Calculate the entropy of the box as an information source.

(2) Let S_0 be the state that Carol is in a good mood, and S_1 be the state that she is in a bad mood. Consider an information source, the output of which is 1 when Carol tells the wrong color and 0 when she tells the correct color. Draw the state transition diagram of this information source.

(3) Consider the steady state after the trial is repeated for a sufficient number of times. Calculate the probability that Carol is in a bad mood.

(4) Calculate the entropy of the information source in Question (2).

(5) Next, consider that Alice informs Bob of the color of the ball through Dave. Suppose that Dave tells Bob the opposite color with the same probability calculated in Question (3). Compare this case with the trial via Carol, and describe the difference in the quantities of information Bob receives. In addition, describe the reason why the difference occurs using the word 'channel capacity.'

Now, suppose that Carol is attached with a sensor that can precisely capture her mood. Consider encoding combinations of her moods in multiple trials by a binary code that consists of zeroes and ones.

(6) Consider assigning codewords for pairs of the moods for two trials. Design the most efficient code. Obtain the average code length per trial in this case.

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(7) Consider increasing the number of trials for combining the moods. Obtain the lower limit of the average code length per trial.

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問題 4

xおよびyを文字'○', '△', '□'から構成される文字列とし，それらをマッチングさせることを考える．まずΩを 3 種の文字の集合とし，2 つの文字$a, b \in \Omega$のスコア$d(a,b)$を$a = b$の時$d(a,b) = 1$，それ以外の時$d(a,b) = -\alpha$とする．ただし$\alpha > 0$とする．文字列xの長さおよびi番目の文字をそれぞれ$|x|$, $x[i]$とする．長さが等しい 2 つの文字列x, yに対し，それらの間のスコアを$D(x,y) := \sum_{i=1}^{|x|} d(x[i], y[i])$と定義する．

次に長さが等しいとは限らない任意の文字列にスコアを拡張する．文字列xの先頭・末尾を含む任意の箇所に任意の個数のギャップ文字'－'を挿入して得られる文字列$\tilde{x}$をxのアライメントと呼ぶ．例えば，$\tilde{x}$ ='－○△－－□－'はx ='○△□'のアライメントの 1 つである．またギャップ文字どうしのスコアを$d(-, -) = -\infty$, ギャップ文字と文字$a \in \Omega$とのスコアを$d(a, -) = d(-, a) = -\beta$とする．ただし$\beta \geq 0$とする．ギャップを含まない 2 つの文字列$x$と$y$の間のアライメントスコア$A(x,y)$とは，$x$と$y$の長さの等しい任意のアライメント$\tilde{x}$と$\tilde{y}$との間のスコア$D(\tilde{x}, \tilde{y})$の中で最大のものとする：

$$A(x,y) := \max_{\tilde{x},\tilde{y}:\, |\tilde{x}|=|\tilde{y}|} D(\tilde{x}, \tilde{y}). \qquad \text{(i)}$$

また最大を達成するアライメントのペアをx^*とy^*とする：

$$\{x^*, y^*\} := \operatorname*{argmax}_{\tilde{x},\tilde{y}:\, |\tilde{x}|=|\tilde{y}|} D(\tilde{x}, \tilde{y}). \qquad \text{(ii)}$$

例えば$\alpha = 1$, $\beta = 1$の時，x ='△□□○□'およびy ='△□△□□□'に対し，$A(x,y)$およびx^*とy^*は$A(x,y) = 2$，x^* ='△□－□○□'およびy^* ='△□△□□□'となる．一般に，x^*とy^*のペアは一意とは限らない．以下の問に答えよ．

(1) $\alpha = 1$, $\beta = 1$とする．この時，x ='□○△'とy ='○□□△'の間のアライメントスコア$A(x,y)$と対応するx^*, y^*とを求めよ．

(2) xとyをそれぞれギャップを含まない長さ$n \geq 1$および$m \geq 1$の文字列とする．スコア$D(\tilde{x}, \tilde{y})$を有限とするアライメント$\tilde{x}$と$\tilde{y}$の組みの個数を$f(n,m)$とする．ただし$f(0,0) = 0$および$f(n,0) = f(0,m) = 1$とする．この時，$f(n,m)$, $f(n-1,m)$, $f(n,m-1)$, $f(n-1,m-1)$の間に成り立つ関係を求めよ．また，$f(3,3)$ の値を求めよ．

(3) ギャップを含まない文字列xに対し，その先頭からk文字を取り出した部分文字列を$x[1:k]$と表す．また$A_{i,j}$を$A_{i,j} = A(x[1:i], y[1:j])$と定義する．$A_{i,j}$を$A_{i-1,j}$, $A_{i,j-1}$, $A_{i-1,j-1}$およびα, βの関数として表わせ．

(4) $A(x,y)$を$O(|x||y|)$の時間計算量で求めるアルゴリズムを擬似コードで示せ．

(5) ギャップを含まない文字列xに含まれるすべての部分文字列の集合を$\sigma(x)$と表す．この

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時，文字列x, yに含まれる任意の部分文字列$s \in \sigma(x)$および$t \in \sigma(y)$の中で，$A(s,t)$を最大にするものがあたえる$A(s,t)$を$L(x,y)$と表す．つまり

$$L(x,y) := \max_{s\in\sigma(x),t\in\sigma(y)} A(s,t), \qquad \text{(iii)}$$

とする．$L(x,y)$を$O(|x||y|)$の時間計算量で求めるアルゴリズムを擬似コードで示せ．

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Problem 4

Suppose x and y be sequences of symbols, '◯', '△', and '□', and let us consider matching of x and y. Let Ω be the set of the three symbols. The score $d(a,b)$ between any two symbols $a, b \in \Omega$ is $d(a,b) = 1$ when $a = b$, and $d(a,b) = -\alpha$ otherwise, where $\alpha > 0$. The length and the ith symbol of a sequence x are denoted by $|x|$ and $x[i]$, respectively. For two sequences x and y of the same length, a score between them is defined as $D(x,y) := \sum_{i=1}^{|x|} d(x[i], y[i])$.

Let us extend the score to sequences, which may have unequal length. Let a sequence $\tilde{x}$ be called an alignment of x. $\tilde{x}$ is obtained by inserting an arbitrary number of gap symbols '$-$' at any positions of x including its head and tail. For example, $\tilde{x}$ ='$-$◯△$--$□$-$' is an alignment of the sequence x ='◯△□'. Let the score between the gap symbols be $d(-, -) = -\infty$, and the score between the gap symbol and a symbol $a \in \Omega$ be $d(a, -) = d(-, a) = -\beta$, where $\beta \geq 0$. An alignment score $A(x,y)$ between two sequences x and y that include no gap symbols, is defined as the maximum among the possible score $D(\tilde{x}, \tilde{y})$ of arbitrary alignments $\tilde{x}$ and $\tilde{y}$ with the same length:

$$A(x,y) := \max_{\tilde{x},\tilde{y};\, |\tilde{x}|=|\tilde{y}|} D(\tilde{x}, \tilde{y}). \qquad \text{(i)}$$

Also, let x^* and y^* be the pair of the alignments that achieve the maximum:

$$x^*, y^* := \operatorname*{argmax}_{\tilde{x},\tilde{y};\, |\tilde{x}|=|\tilde{y}|} D(\tilde{x}, \tilde{y}). \qquad \text{(ii)}$$

For example, if $\alpha = 1$ and $\beta = 1$, the alignment score $A(x,y)$, x^*, and y^* of x ='△□□◯□' and y ='△□△□□□' are $A(x,y) = 2$, x^* ='△□$-$□◯□', and y^* ='△□△□□□', respectively. The pair x^* and y^* may not be unique in general. Answer the following questions.

(1) Suppose that $\alpha = 1$ and $\beta = 1$. Obtain the alignment score $A(x,y)$ and the corresponding x^* and y^* for x ='□◯△' and y ='◯□□△'.

(2) Suppose x and y be sequences without the gap symbols of length $n \geq 1$ and $m \geq 1$, respectively. Let $f(n,m)$ be the number of pairs of alignments $\tilde{x}$ and $\tilde{y}$, the score $D(\tilde{x}, \tilde{y})$ of which is finite, where we set $f(0,0) = 0$ and $f(n,0) = f(0,m) = 1$. Derive a relation that holds between $f(n,m)$, $f(n-1,m)$, $f(n,m-1)$, and $f(n-1,m-1)$. Also, calculate $f(3,3)$.

(3) Suppose x be a sequence including no gap symbol and $x[1:k]$ be the sequence of the first k symbols of x. Also define $A_{i,j}$ by $A_{i,j} = A(x[1:i], y[1:j])$. Describe $A_{i,j}$ as a function of $A_{i-1,j}$, $A_{i,j-1}$, $A_{i-1,j-1}$, α, and β.

(4) Write a pseudo code of an algorithm that calculates $A(x,y)$, and has the time complexity of $O(|x||y|)$.

(5) Let x be a sequence that includes no gap symbols. Next, let $\sigma(x)$ be the set of all subsequences

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of x. For two sequences x and y, we define $L(x,y)$ as the maximum of $A(s,t)$ where $s \in \sigma(x)$ and $t \in \sigma(y)$. Namely,

$$L(x,y) := \max_{s\in\sigma(x),t\in\sigma(y)} A(s,t). \qquad \text{(iii)}$$

Write a pseudo code of an algorithm that calculates $L(x,y)$, and has the time complexity of $O(|x||y|)$.

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問題 A

図 1 の回路をQ_0の初期値を1，$Q_1,\ldots,Q_{n-1}$の初期値を0として動作させ，0，1がおおよそ同じ割合で出現する1ビットの擬似乱数を1クロックごとに発生させることを考える．ただしnは 2 以上の整数とする．以下の問に答えよ．

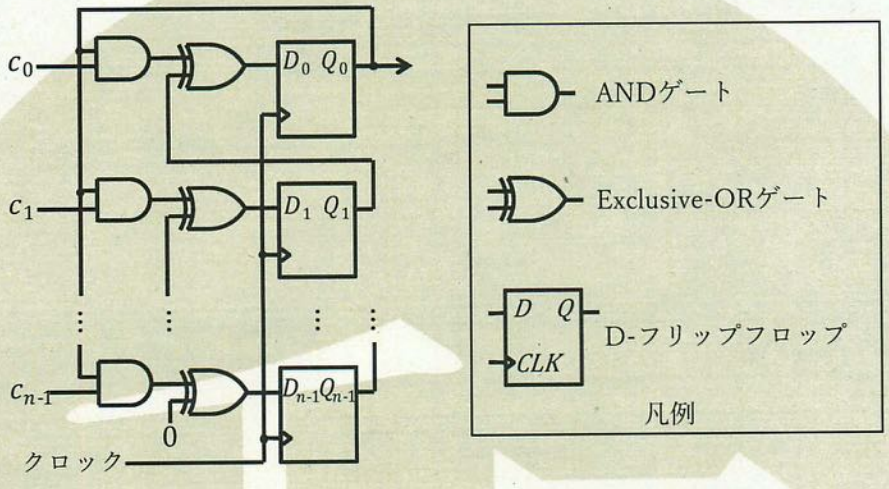


図 1

(1) 回路の動作中，入力$c_0,\ldots,c_{n-1}$が変化しない場合，出力Q_0は周期性を持つ．$n=4$，$c_0=c_1=c_2=c_3=1$の場合の周期を求めよ．

(2) $n=4$で，回路の動作中，常に入力$c_0=c_1=0$，$c_2=c_3=1$である場合の周期と，1周期分の出力Q_0の時系列を示せ．

(3) 回路の動作中，入力$c_0,\ldots,c_{n-1}$が変化しない場合，出力Q_0の周期は高々2^n-1であり，また，周期が2^n-1であるためには$c_{n-1}=1$であることが必要である．その理由を説明せよ．

(4) 問(2)の状況において，ある時点において出力Q_0がi $(i=0,1)$であるときに，次のクロックにおける出力Q_0がj $(j=0,1)$である条件付き確率p_{ij}を求め，理想的な乱数の場合と比較せよ．

(5) 問(2)の状況において，ある時点における出力Q_0が0であるときに，その後ちょうどkクロックの間だけ連続して出力Q_0が1となる条件付き確率をq_{0k} $(k=1,2,\ldots)$，ある時点における出力Q_0が1であるときに，その後ちょうどkクロックの間だけ連続して出力Q_0が0となる条件付き確率をq_{1k} $(k=1,2,\ldots)$とする．q_{0k}とq_{1k}を求め，理想的な乱数の場合と比較せよ．

(6) 出力Q_0をより真の乱数に近づけるため，nの値を変えずに図 1 の回路に変更を加えて，出力Q_0の周期を2^n-1より大きくするためには，どのような方策が考えられるか述べよ．

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(7) 擬似乱数ではない，真の乱数の生成器について，知るところを数行で述べよ．

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Problem A

Suppose that the circuit shown in Fig. 1 with 1 as the initial value of Q_0 and 0 as the initial values of $Q_1, \ldots, Q_{n-1}$, is used to generate a 1-bit pseudo random number with nearly equal appearance probability for 0 and 1 at each clock. Note that n is an integer and $n \geq 2$. Answer the following questions.

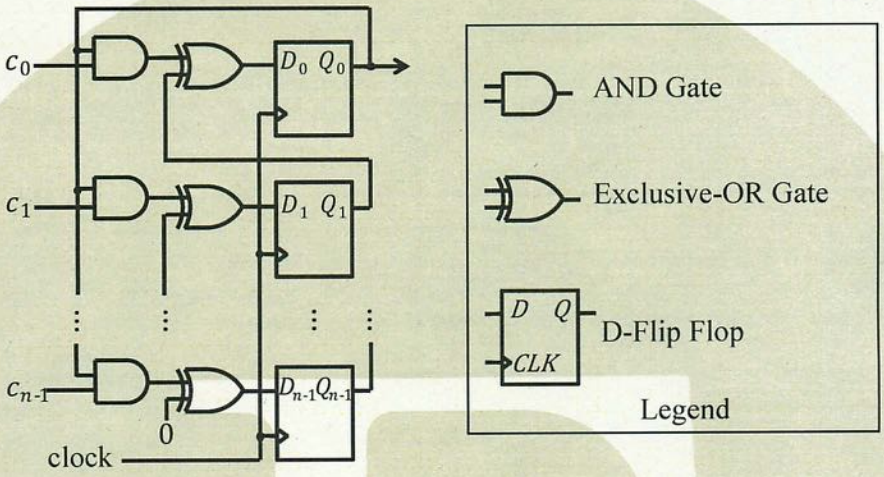


Fig. 1

(1) If the inputs $c_0, \ldots, c_{n-1}$ do not change during the operations of the circuit, the output Q_0 becomes periodic. Obtain the period for the case that $n = 4$ and $c_0 = c_1 = c_2 = c_3 = 1$.

(2) Obtain the period and the one period time sequence of the output Q_0 for the case that $n = 4$, the inputs $c_0 = c_1 = 0$ and $c_2 = c_3 = 1$.

(3) If the inputs $c_0, \ldots, c_{n-1}$ do not change during operation, the period of the output Q_0 does not exceed $2^n - 1$, and $c_{n-1} = 1$ is required for the period to be $2^n - 1$. Explain why ths is so.

(4) In the case described in Question (2), calculate the conditional probabilities p_{ij} that the output Q_0 becomes j $(j = 0, 1)$ at a certain clock under the condition that the output Q_0 was i $(i = 0, 1)$ at the previous clock. Compare the probabilities with those of the ideal random number case.

(5) In the case described in Question (2), let q_{0k} $(k = 1, 2, \ldots)$ be the conditional probability that the output Q_0 becomes 1 for exactly and only k consecutive clock periods following the clock when the output Q_0 was 0, and q_{1k} $(k = 1, 2, \ldots)$ be the conditional probability that the output Q_0 becomes 0 for exactly and only k consecutive clock periods following the clock when the output Q_0 was 1. Calculate q_{0k} and q_{1k} and compare them with those of the ideal random number case.

(6) Give an idea of how to make the periodic interval of the output Q_0 larger than $2^n - 1$ by

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modifying the circuit shown in Fig. 1 without changing n, in order to make the output Q_0 look more like a true random number.

(7) Write your knowledge about generators of true random numbers instead of pseudo random numbers in a few lines.

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問題 B

図 1 に示すように，パルス源からのパルスをローパスフィルタ(LPF)に入力し，出力パルスを得るシステムを考える．パルス源は周期Tのパルス列を発生し，そのパルスの幅は非常に短くデルタ関数と見なせるものとする．以下の問に答えよ．

(1) LPF が理想的な矩形 LPF，すなわちその周波数特性$H_r(f)$が図 2 に示すように$|f| \leq B$（透過域）で$H_r(f) = 1$，$|f| > B$（阻止域）で$H_r(f) = 0$であるとする．単一のパルスを入力したときの出力パルス波形の式を求め，その概形を図示せよ．

(2) このシステムからの出力パルス列を用いて，パルスの振幅に情報を乗せるパルス振幅変調(PAM)によりデータ伝送を行いたい．ただし，LPF は問(1)の矩形 LPF であるとする．
  (a) 個々の出力パルスのピークが他のパルスにより干渉を受けないようにするには，周波数Bをどのように選べばよいか．
  (b) 問(2)(a)での条件を満たすBのうち最小の周波数$B_{\min}$を求めよ．

(3) 矩形 LPF に代えて，図 3 に示すような透過域$|f| \leq B_1$と阻止域$|f| > B_2$の間に直線の遷移域$B_1 < |f| \leq B_2$を設けた台形 LPF を用いることにする．ただし，$B_2 - B_1$は問(2)(b)で求めた$B_{\min}$より小さいとする．
  (a) この台形 LPF の周波数特性$H_t(f)$は，図 4 の 2 つの矩形 LPF の周波数特性$H_a(f), H_b(f)$の畳み込みで表されることを証明せよ．
  (b) この台形 LPF に単一のパルスを入力したときの出力パルス波形の式を求め，その概形を図示せよ．
  (c) この台形 LPF を用いた PAM 伝送について，問(2)(b)と同様に個々の出力パルスのピークが他のパルスにより干渉を受けないようにし，かつ透過域を最小にしたい．このときにB_1, B_2が満たすべき条件を示せ．
  (d) PAM 伝送に用いる LPF として，矩形 LPF とこの台形 LPF とを比較してその長所・短所を述べよ．

(4) 上述の矩形 LPF と台形 LPF に線形な位相特性を与える，つまりそれぞれの周波数特性が$H_r(f)\exp(-j2\pi f\tau_r)$，$H_t(f)\exp(-j2\pi f\tau_t)$と表される場合を考える．
  (a) このときには問(1)および問(3)(b)で得たパルス波形はどう変化するか．
  (b) 問(1)および問(3)で考えたような LPF は現実には作れない．その理由を述べるとともに，どのような位相特性を与えれば現実的な LPF が作れるかを述べよ．

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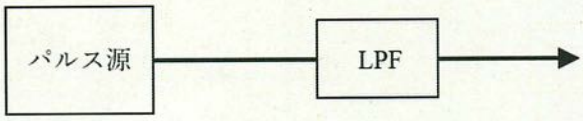


図 1

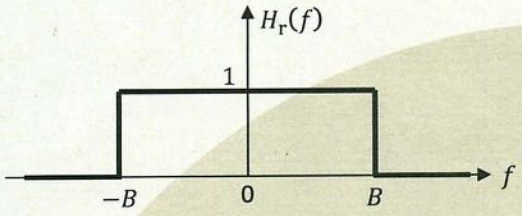


図 2

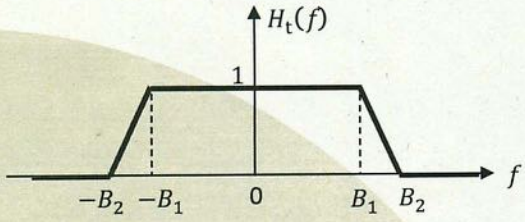
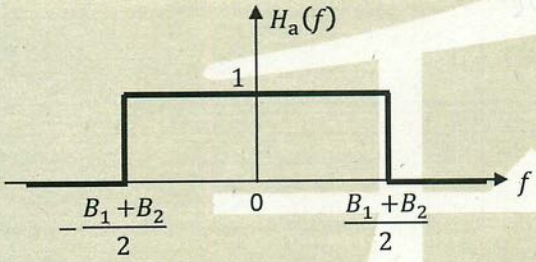


図 3



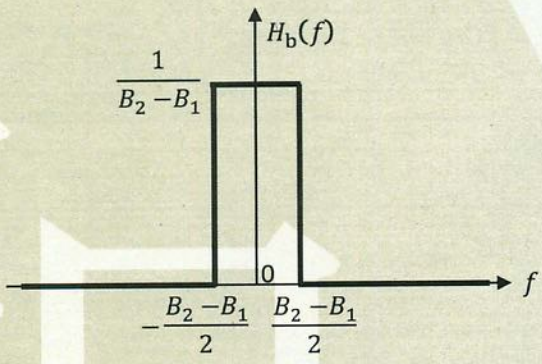


図 4

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Problem B

Consider a system in which pulses from a pulse source pass through a low-pass filter (LPF), as shown in Fig. 1. The pulse source generates a pulse train whose period is T, and each pulse is short enough to be considered as the delta function. Answer the following questions.

(1) Assume that the LPF is an ideal rectangular LPF, that is, its frequency response $H_r(f)$ is given as $H_r(f) = 1$ for $|f| \leq B$ (passband) and $H_r(f) = 0$ for $|f| > B$ (stopband), as shown in Fig. 2. Obtain a formula of the output pulse waveform from this system when a single pulse is input to the LPF, and sketch the outline of the waveform.

(2) Consider a data transmission using the output pulse train from this system based on pulse amplitude modulation (PAM) in which the information is encoded on the pulse amplitudes. Here the LPF is assumed to be the ideal rectangular LPF from Question (1).
   (a) How should the frequency B be chosen such that each pulse peak does not interfere with the other pulses?
   (b) Obtain $B_{\min}$, the minimum frequency satisfying the condition of Question (2)(a).

(3) We replace the rectangular LPF with a trapezoidal LPF having a linear transition region $B_1 < |f| \leq B_2$ in between the passband $|f| \leq B_1$ and the stopband $|f| > B_2$, as shown in Fig. 3. Here $B_2 - B_1$ is assumed to be smaller than $B_{\min}$ obtained in Question (2)(b).
   (a) Prove that the frequency response $H_t(f)$ of the trapezoidal LPF is given by the convolution of frequency responses $H_a(f)$ and $H_b(f)$ of two rectangular LPFs shown in Fig. 4.
   (b) Obtain a formula of the output pulse waveform when a single pulse is input to the trapezoidal LPF, and sketch the outline of the waveform.
   (c) We design the PAM data transmission so that each pulse peak does not interfere with the other pulses, and the passband is minimum, as was in Question (2)(b). Find a condition that B_1 and B_2 have to satisfy.
   (d) Compare the rectangular LPF and trapezoidal LPF in terms of the PAM data transmission, and describe their advantages and disadvantages.

(4) We add the linear phase responses to the above rectangular LPF and trapezoidal LPF, that is, respective frequency responses are given as $H_r(f)\exp(-j2\pi f\tau_r)$ and $H_t(f)\exp(-j2\pi f\tau_t)$.
   (a) How are the pulse waveforms obtained in Questions (1) and (3)(b) changed?
   (b) The LPFs in Questions (1) and (3) are not realizable. Describe reasons, and discuss what kind of phase responses are needed for the LPFs to be realizable.

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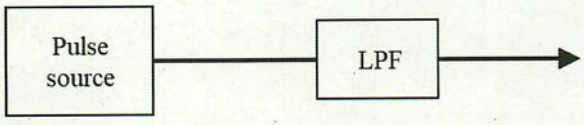


Fig. 1

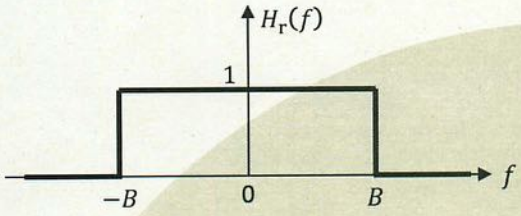


Fig. 2

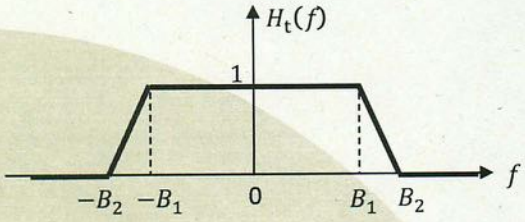
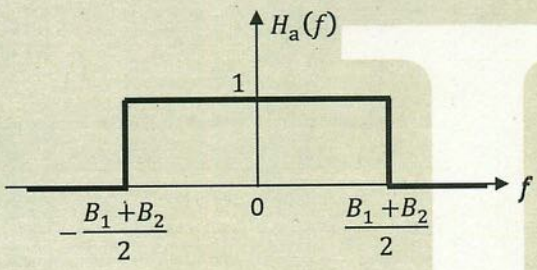


Fig. 3



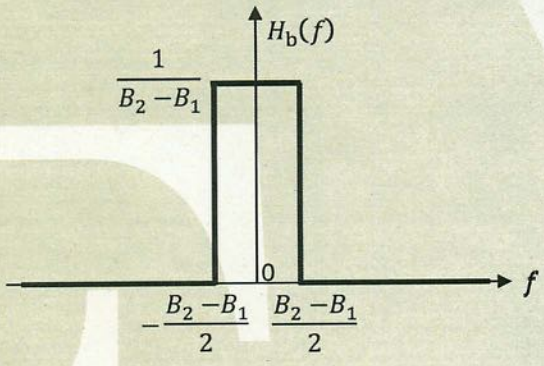


Fig. 4

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問題 C

量子トンネル効果や量子井戸を，電子に対して式(i)で表されるシュレディンガー方程式を使って考える．

$$i\hbar\frac{\partial}{\partial t}\psi(x,t)=\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+V(x)\right]\psi(x,t) \qquad \text{(i)}$$

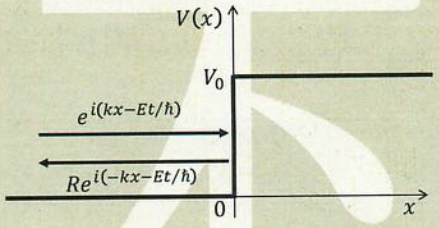
ただし，xは位置，tは時刻，$\psi(x,t)$は電子の波動関数，iは虚数単位，$\hbar=h/(2\pi)$であり，hはプランク定数，mは電子の質量，$V(x)$はポテンシャルとする．以下の問に答えよ．

(1) $V(x)=V_0$で一定のポテンシャルを考える．このとき，エネルギーE，波数kを持ち，一方向に伝搬する電子の波動関数は，式(ii)のように表される．

$$\psi(x,t)=Ce^{i(kx-Et/\hbar)} \qquad \text{(ii)}$$

ただし，Cは定数である．このとき，Eとkに関する分散関係を求めよ．

(2) 図1に示すステップ状ポテンシャルに，エネルギーEの電子が左から入射する場合を考える．ただし，$0<E<V_0$とする．このとき，$x<0$における波動関数は入射波$e^{i(kx-Et/\hbar)}$と反射波$Re^{i(-kx-Et/\hbar)}$の和で表され，$x\geq 0$に染み出す波動関数は$C_1e^{i(k_1x-Et/\hbar)}$と表される．このとき，k_1およびC_1をE，V_0およびmを使って表せ．ただし，Rは定数である．

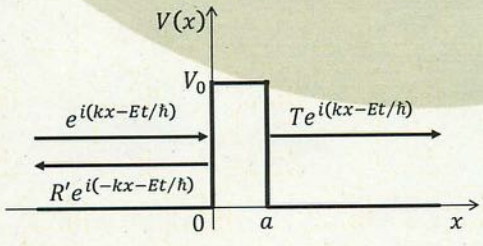


$$V(x)=\begin{cases}V_0 & (x\geq 0)\\ 0 & (x<0)\end{cases}$$

ただし，V_0は定数（$V_0>0$）

図1

(3) 図2に示す厚さaのポテンシャル障壁に，エネルギー$E=V_0/2$の電子が左から入射する場合を考える．このとき，量子トンネル効果により，電子が障壁を通り抜ける可能性がある．$x<0$における波動関数は入射波$e^{i(kx-Et/\hbar)}$と反射波$R'e^{i(-kx-Et/\hbar)}$の和で表され，$x>a$での波動関数は$Te^{i(kx-Et/\hbar)}$と表される．TをV_0，mおよびaを使って表せ．ただし，R'は定数である．



$$V(x)=\begin{cases}0 & (x>a)\\ V_0 & (0\leq x\leq a)\\ 0 & (x<0)\end{cases}$$

ただし，V_0は定数（$V_0>0$）

図2

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(4) 図3に示す厚さwの量子井戸を考える．量子井戸に閉じ込められた電子の定常状態は，図3に示すように離散的なエネルギーE_1，E_2，…，を持つ．$V_0 \to \infty$としたときの電子の取りうる最小エネルギーE_1および，そのときの波動関数を求めよ．

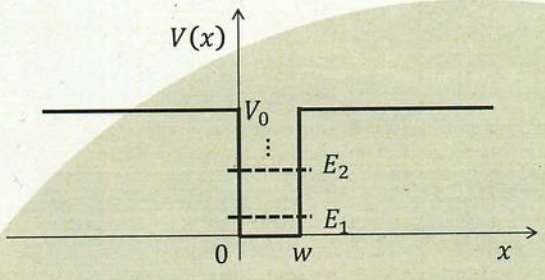


図3

$$V(x) = \begin{cases} V_0 & (x > w) \\ 0 & (0 \le x \le w) \\ V_0 & (x < 0) \end{cases}$$

ただし，V_0は定数（$V_0 > 0$）

(5) 2つの同一の量子井戸を考える．このとき，最低エネルギーE_1を持つそれぞれの量子井戸中の電子の波動関数を$\varphi_1 e^{-iE_1 t/\hbar}$，$\varphi_2 e^{-iE_1 t/\hbar}$とする．2つの量子井戸を図4に示すように近接させると，量子トンネル効果により，2つの量子井戸間をある確率で電子が移動することができる．このとき，系全体の波動関数は，$\psi(x,t) = c_1(t)\varphi_1 + c_2(t)\varphi_2$と表される．$c_1$および$c_2$は式(iii)，(iv)を満たす．

$$i\hbar \frac{dc_1}{dt} = E_1 c_1 - A c_2 \qquad \text{(iii)}$$

$$i\hbar \frac{dc_2}{dt} = E_1 c_2 - A c_1 \qquad \text{(iv)}$$

ただし，Aは定数とする．このとき系全体の定常状態におけるエネルギーを求めよ．

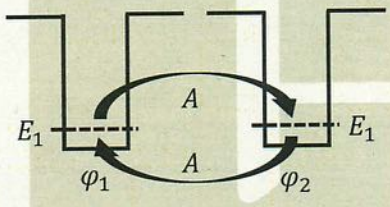


図4

(6) 固体を，量子井戸が一列に無数に並んだモデルで近似することを考える．問(5)の結果を用いて，固体中のバンド構造について図も使って10行以内で定性的に述べよ．

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Problem C

Consider quantum tunneling and a quantum well using the Schrödinger equation for an electron given by Eq. (i).

$$i\hbar\frac{\partial}{\partial t}\psi(x,t)=\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+V(x)\right]\psi(x,t), \tag{i}$$

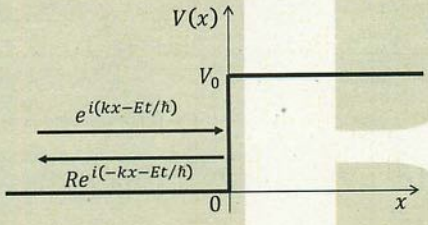
where x is position, t is time, $\psi(x,t)$ is the wave function for an electron, i is the imaginary unit, $\hbar=h/(2\pi)$, h is the Planck constant, m is the electron mass, and $V(x)$ is the potential. Answer the following questions.

(1) Assume a constant potential given by $V(x)=V_0$. The wave function for an electron with an energy E and a wave number k propagating in one direction is expressed by Eq. (ii).

$$\psi(x,t)=Ce^{i(kx-Et/\hbar)}, \tag{ii}$$

where C is a constant. Show the dispersion relationship between E and k.

(2) Consider an electron with energy E coming in from the left into the potential step shown in Fig. 1, where $0<E<V_0$. The wave function in $x<0$ is expressed by the sum of an incident wave $e^{i(kx-Et/\hbar)}$ and a reflected wave $Re^{i(-kx-Et/\hbar)}$, and the wave function penetrating into the region $x\geq 0$ is expressed by $C_1e^{i(k_1x-Et/\hbar)}$. Express k_1 and C_1 by E, V_0, and m. Note that R is a constant.

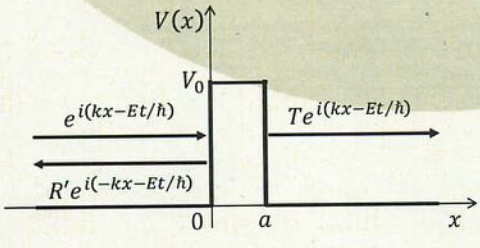


$$V(x)=\begin{cases}V_0 & (x\geq 0)\\ 0 & (x<0)\end{cases}$$

where V_0 is a constant ($V_0>0$).

Fig. 1

(3) Consider an electron with an energy $E=V_0/2$ coming in from the left into the potential barrier with a thickness of a shown in Fig. 2. Due to quantum tunneling, the injected electron tunnels the barrier with a certain probability. The wave function in $x<0$ is expressed by the sum of an incident wave $e^{i(kx-Et/\hbar)}$ and a reflected wave $R'e^{i(-kx-Et/\hbar)}$, and the wave function in $x>a$ is expressed by $Te^{i(kx-Et/\hbar)}$. Express T by V_0, m, and a. Note that R' is a constant.



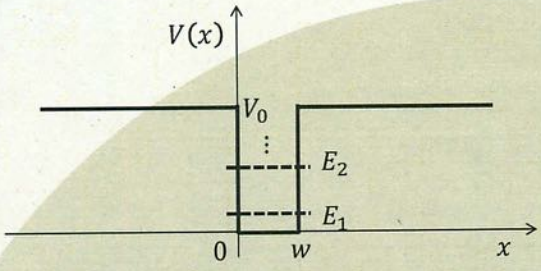
$$V(x)=\begin{cases}0 & (x>a)\\ V_0 & (0\leq x\leq a)\\ 0 & (x<0)\end{cases}$$

where V_0 is a constant ($V_0>0$).

Fig. 2

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(4) Consider a quantum well with a thickness of w shown in Fig. 3. The steady states of an electron confined in the quantum well have discrete energies E_1, E_2, $\cdots$ as shown in Fig. 3. Assuming $V_0 \to \infty$, derive the lowest energy E_1 and the corresponding wave function.



$$V(x) = \begin{cases} V_0 & (x > w) \\ 0 & (0 \le x \le w) \\ V_0 & (x < 0) \end{cases}$$

where V_0 is a constant $(V_0 > 0)$.

Fig. 3

(5) Consider two identical quantum wells. Assume that the wave functions for an electron in the respective quantum wells at the lowest energy E_1 are $\varphi_1 e^{-iE_1 t/\hbar}$ and $\varphi_2 e^{-iE_1 t/\hbar}$. When the two quantum wells are close together as shown in Fig. 4, an electron can move between the two quantum wells with a certain probability due to quantum tunneling. In this situation, the wave function of the whole system is expressed by $\psi(x,t) = c_1(t)\varphi_1 + c_2(t)\varphi_2$. $c_1(t)$ and $c_2(t)$ satisfy Eqs. (iii) and (iv).

$$i\hbar \frac{dc_1}{dt} = E_1 c_1 - A c_2, \tag{iii}$$

$$i\hbar \frac{dc_2}{dt} = E_1 c_2 - A c_1, \tag{iv}$$

where A is a constant. Calculate the energies of the whole system when in the steady state.

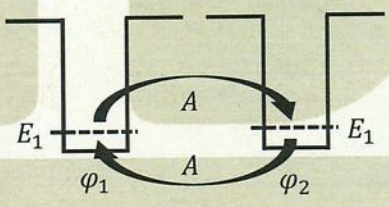


Fig. 4

(6) Consider an approximate model for a solid using numerous quantum wells in a line. By using the result of Question (5), explain a band structure of a solid qualitatively in 10 lines with figure(s).

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問題 D

図 1 に示す n 型の金属・酸化物・半導体電界効果トランジスタ（MOSFET）を考える．ソースと基板は接地されている．ゲート電圧V_Gが一定で$V_G > V_T$の時，この MOSFET は図 2 の$I_D - V_D$特性を示す．ただし，I_Dはドレイン電流を，V_Dはドレイン電圧を示す．図 2 の領域 1 の$I_D - V_D$特性は式(i)で表される．

$$I_D = \mu C_{OX} \frac{W}{L}\left\{(V_G - V_T)V_D - \frac{1}{2}{V_D}^2\right\} \qquad \text{(i)}$$

ただし，μはチャネル中の電子移動度，C_{OX}は単位面積当たりのゲート酸化膜容量，Lはチャネル長，Wはチャネル幅である．チャネル長は十分に長く，μとV_Tは一定と仮定する．以下の問に答えよ．

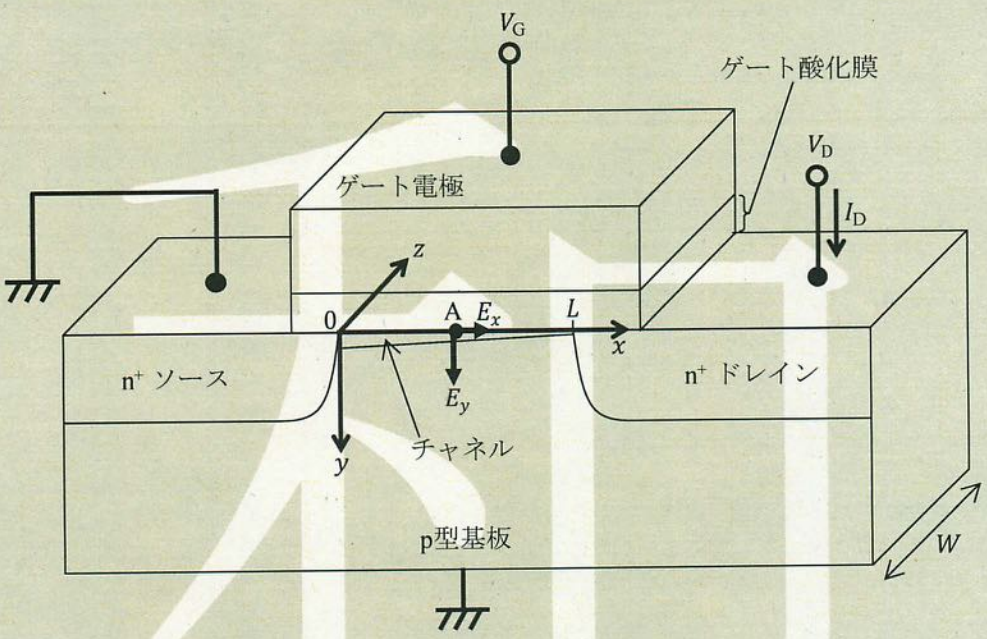


図 1

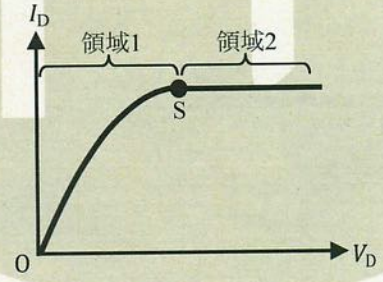


図 2

(1) 式(i)のV_Tの名称を答えよ．V_Tの意味を 1 行で説明せよ．

(2) 図 2 の領域 1 と領域 2 のそれぞれの名称を答えよ．

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(3) 式(i)の導出を考える．図 1 に示すようにxyz座標を定義する．y方向電界(E_y)はx方向電界(E_x)よりも十分に大きいと仮定する．基板表面($y = 0$)の点 A のチャネル電位を$V_C(x)$と定義する．ソース端($x = 0$)において，$V_C(0) = 0$Vである．ドレイン端($x = L$)において，$V_C(L) = V_D$である．点 A での単位ゲート面積あたりの反転電荷密度$Q(x)$は次のように表される．

$$|Q(x)| = C_{OX}\{V_G - V_C(x) - V_T\} \qquad \text{(ii)}$$

I_Dはチャネル中で一定である．I_Dの導出においてドリフト電流のみを考慮せよ．チャネル中の電子のドリフト速度の大きさ$|v|$は$\mu|E_x|$に等しい．

(3-1) I_Dを$|Q(x)|$と$|v|$を用いて表せ．

(3-2) $|E_x|$を$V_C(x)$を用いて表せ．

(3-3) 問(3-1)と問(3-2)の結果を用いて式(i)を導出せよ．

(4) 図 2 の点 S のV_Dを求めよ．

(5) 図 2 の領域 2 でI_Dが一定になる理由を図と 2〜3 行の文章で説明せよ．

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Problem D

Consider an n-type metal-oxide-semiconductor field-effect transistor (MOSFET) shown in Fig. 1. The source and the substrate are connected to the ground. When the gate voltage V_G is constant and $V_G > V_T$, the MOSFET exhibits the $I_D - V_D$ characteristic shown in Fig. 2, where I_D is the drain current and V_D is the drain voltage. The $I_D - V_D$ characteristic of Region 1 in Fig. 2 is represented by Eq. (i).

$$I_D = \mu C_{OX} \frac{W}{L} \left\{ (V_G - V_T) V_D - \frac{1}{2} {V_D}^2 \right\}, \qquad \text{(i)}$$

where μ is the electron mobility in the channel, C_{OX} is the gate oxide capacitance per unit area, L is the channel length, and W is the channel width. Assume that the channel length is sufficiently long, and that μ and V_T are constant. Answer the following questions.

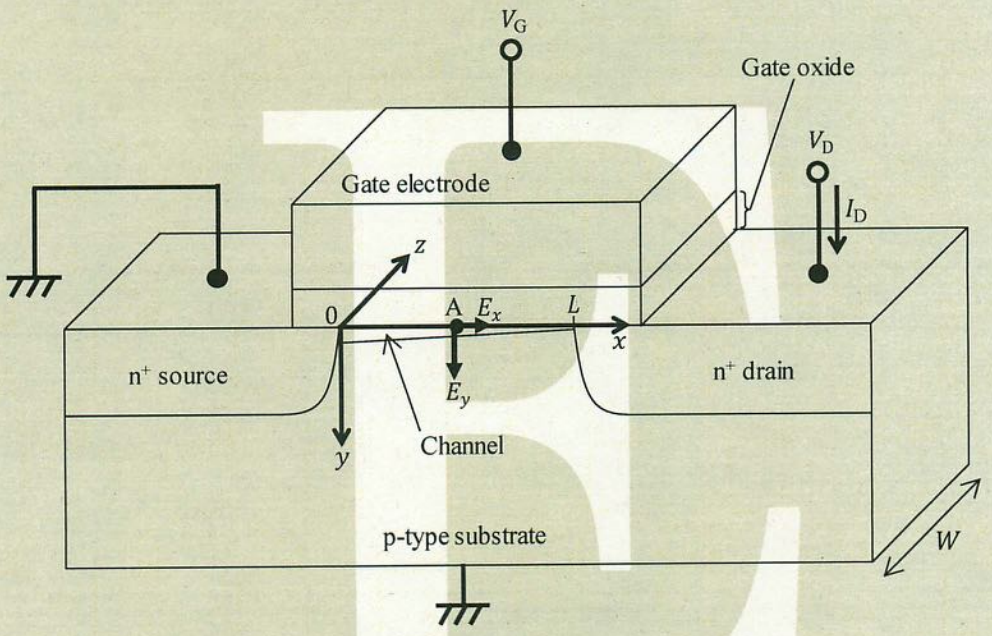


Fig. 1

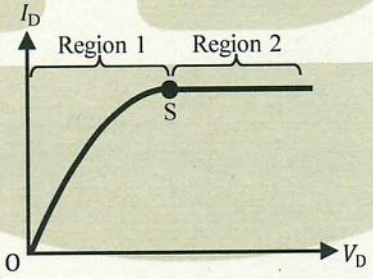


Fig. 2

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(1) What is the common name of V_T in Eq. (i) ? Explain the meaning of V_T in one line.

(2) What are Region 1 and Region 2 in Fig. 2 usually called ?

(3) Consider the derivation of Eq. (i). The xyz coordinates are defined as shown in Fig. 1. Assume that the electric field in the y direction (E_y) is much larger than the electric field in the x direction (E_x). The channel potential at a point A on the surface of the substrate ($y = 0$) is defined as $V_C(x)$. At the source edge ($x = 0$), $V_C(0) = 0$V. At the drain edge ($x = L$), $V_C(L) = V_D$. The inversion charge density per unit gate area at the point A, $Q(x)$, is given by

$$|Q(x)| = C_{OX}\{V_G - V_C(x) - V_T\}. \qquad \text{(ii)}$$

I_D is constant in the channel. In the derivation of I_D, consider only drift current. The magnitude $|v|$ of the drift velocity of the electrons in the channel is equal to $\mu|E_x|$.

(3-1) First, express I_D with $|Q(x)|$ and $|v|$.

(3-2) Next, express $|E_x|$ with $V_C(x)$.

(3-3) Finally, derive Eq. (i) by using the results of Question (3-1) and Question (3-2).

(4) Find V_D of the point S in Fig. 2.

(5) Explain the reason why I_D is constant in Region 2 of Fig. 2 in two or three lines using figures.

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問題E

制御対象および制御器の伝達関数$P(s)$, $C(s)$が下式で与えられるフィードバック制御系がある．この制御器$C(s)$の微分動作の効果を調べる．

$$P(s) = \frac{1}{s(s+1)(s+2)}, \qquad C(s) = K(1+Ts).$$

以下の問に答えよ．

(1) $T = 0$とし，Kを0から∞まで変化させて，閉ループ系の根軌跡の概形を描け．

(2) 問(1)で安定限界を与えるKを求めよ．

(3) Kを問(2)で求めた値に固定し，Tを0から∞まで変化させた場合の閉ループ系の根軌跡の概形を描け．閉ループ系の特性多項式を，Tを含む部分と含まない部分に分け，Tがある仮想的な制御対象を制御する比例制御器のゲインとなるようなフィードバック制御系を考えるとよい．

(4) 問(3)の結果をもとに，微分動作の効果について数行で述べよ．

(5) 問(2)で求めたK以外のいくつかのKについても根軌跡を描き，問(4)で答えた効果の限界について数行で述べよ．

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Problem E

The transfer functions of the plant and the controller $P(s)$ and $C(s)$ are given as

$$P(s) = \frac{1}{s(s+1)(s+2)} \text{ and } C(s) = K(1+Ts),$$

respectively. Investigate the effect of the derivative operation of the controller $C(s)$ in a closed loop feedback system. Answer the following questions.

(1) Let $T = 0$ and draw a rough sketch of the root locus of the closed loop system by changing K from 0 to ∞.

(2) Find K that gives the stability limit in Question (1).

(3) Draw a rough sketch of the root locus of the closed loop system where K is fixed to the value found in Question (2) and T is changed from 0 to ∞. It is recommended to divide the characteristic polynomial of the closed loop system into a part including T and a part not including T, and to consider a proportional feedback system that controls a virtual plant with a proportional gain of T.

(4) Describe the effect of the derivative operation based on the results obtained in Question (3) in a few lines.

(5) Draw a rough sketch of the root loci for K's other than the one derived in Question (2), and describe the limitation of the effect answered in Question (4) in a few lines.

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問題 F

電力工学に関する以下の問に答えよ．ただし，$\dot{V}$，$\dot{I}$は正弦波交流電圧，電流の複素数表示であり，その絶対値は実効値を表す．jは虚数単位を表す．送電線の線路インピーダンスの抵抗成分をR，リアクタンス成分をXとする．負荷のインピーダンスを$\dot{Z}_L$とする．

(1) 等価回路が図 1 で表現できる送電系統を考える．このとき，電流$\dot{I}_0$，送電端電圧$\dot{V}_S$，受電端電圧$\dot{V}_R$の関係をベクトル図（フェーザ図）に示せ．負荷の力率角φを図中に示せ．
(2) この送電系統の送電端における有効電力と無効電力を，$|\dot{I}_0|$，R，X，$|\dot{Z}_L|$，φのうち必要なものを用いて示せ．その導出の過程も示せ．
(3) この送電系統の電圧降下$|\dot{V}_S| - |\dot{V}_R|$を，$|\dot{I}_0|$，$R$，$X$，$\varphi$のうち必要なものを用いて示せ．その導出の過程も示せ．ただし，$R \ll |\dot{Z}_L|$，$X \ll |\dot{Z}_L|$とする．

図 2 のように，交流送配電方式には (a)単相 2 線式，(b)単相 3 線式，(c)三相 4 線式などがある．

(4) 図 2 の各方式において，線電流の最大値，相電圧，および負荷の力率が等しいとき，(a)の方式の総送電電力を 1 とした場合の(b)，(c)の方式それぞれの総送電電力を示せ．ただし，どの方式も負荷および交流電源は平衡しているものとし，$R \ll |\dot{Z}_L|$，$X \ll |\dot{Z}_L|$であるとする．
(5) 図 2 の各方式において，総送電電力，相電圧，および負荷の力率が等しいとき，(a)の方式の送電線の総電力損失を 1 とした場合の(b)，(c)の方式それぞれの送電線の総電力損失を示せ．ただし，どの方式も負荷および交流電源は平衡しているものとし，$R \ll |\dot{Z}_L|$，$X \ll |\dot{Z}_L|$，電力損失はR部分のみで生ずるものとする．
(6) 実際の送配電線では上記の理想的な平衡条件は成り立たない．このことを考慮して，送配電方式(a)，(b)，(c)はどのような場合に使用されているか，その理由とともに，それぞれ数行で示せ．

本サンプルは2020年度以前の問題に基づいたものであり、問題形式、分量、出題範囲、難易度において現在のものと異なります。
This sample is based on the problems before FY2020, which are different from the current ones in their style, volume, range, and level.

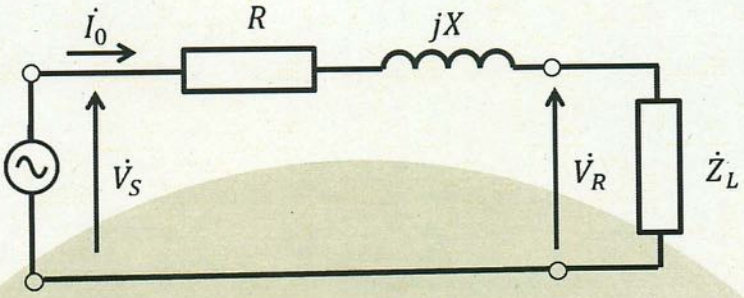
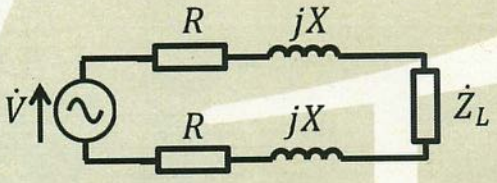
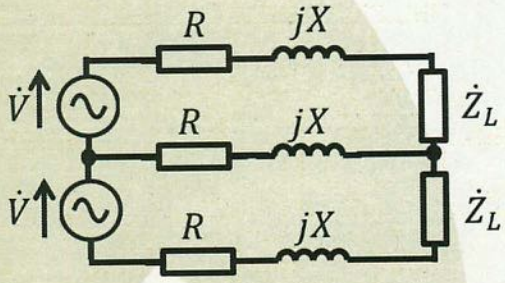


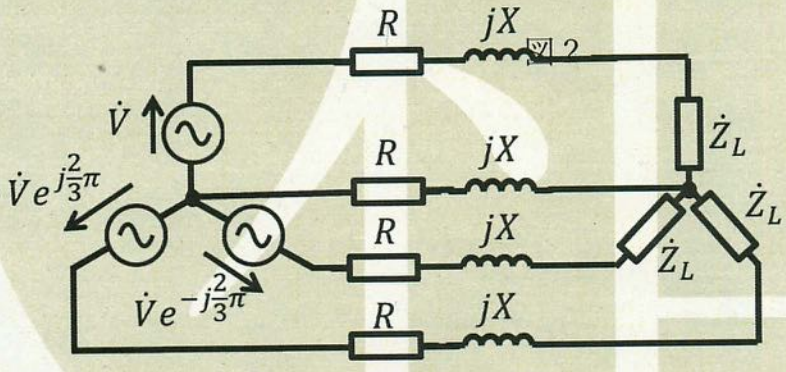
図 1



(a) 単相2線式



(b) 単相3線式



(c) 三相4線式

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Problem F

Answer the following questions with regard to power engineering. Note that $\dot{V}$ and $\dot{I}$ denote the complex number expressions of sinusoidal AC voltage and current, respectively, and their absolute values correspond to effective values. Note also that j denotes the imaginary unit. The line impedance of each power transmission line has a resistance component R and a reactance component X. The load impedance is $\dot{Z}_L$.

(1) Let us consider a power transmission system shown in Fig. 1. Draw the relation between the current $\dot{I}_0$, the voltage at the sending end $\dot{V}_S$, and the voltage at the receiving end $\dot{V}_R$ in a vector diagram (phasor diagram). Indicate the power factor angle φ of the load in the diagram.

(2) Find the active power and reactive power at the sending end by using some of $|\dot{I}_0|$, R, X, $|\dot{Z}_L|$, and φ. Show your derivation process.

(3) Find the voltage drop of the power transmission system $|\dot{V}_S| - |\dot{V}_R|$ by using some of $|\dot{I}_0|$, R, X, and φ, and show your derivation process. Note that $R \ll |\dot{Z}_L|$ and $X \ll |\dot{Z}_L|$.

As shown in Fig. 2, there are some types of AC power distribution systems such as (a) single-phase two-wire system, (b) single-phase three-wire system, and (c) three-phase four-wire system.

(4) Under the conditions that the maximum value of line currents, the phase voltages, and the power factors of the loads of every power distribution system in Fig. 2 are the same, find the total transmitting powers of systems (b) and (c) assuming that the total transmitting power of system (a) is 1. Note that loads and AC power sources are balanced in every system, $R \ll |\dot{Z}_L|$, and $X \ll |\dot{Z}_L|$.

(5) Under the conditions that the total transmitting powers, the phase voltages, and the power factors of the loads of every power distribution system in Fig. 2 are the same, find the total power losses of systems (b) and (c) assuming that the power loss of system (a) is 1. Note that the loads and AC power sources are balanced in every system, $R \ll |\dot{Z}_L|$, $X \ll |\dot{Z}_L|$, and that the power loss occurs only in the resistance R.

(6) In a real system, the ideal balancing assumptions mentioned above will not be satisfied. Considering this fact, explain practical applications of power distribution systems (a), (b), and (c) with reasons in a few lines for each system.

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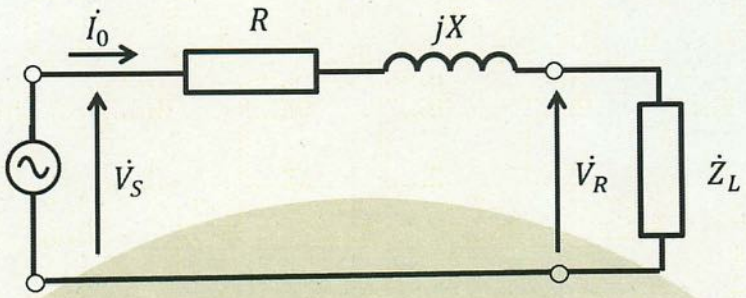
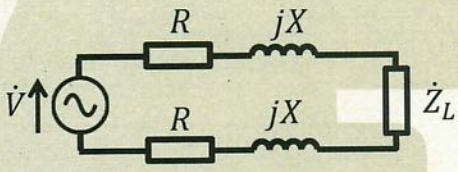
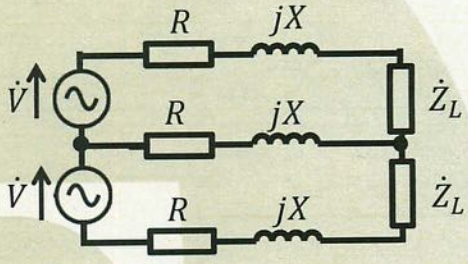


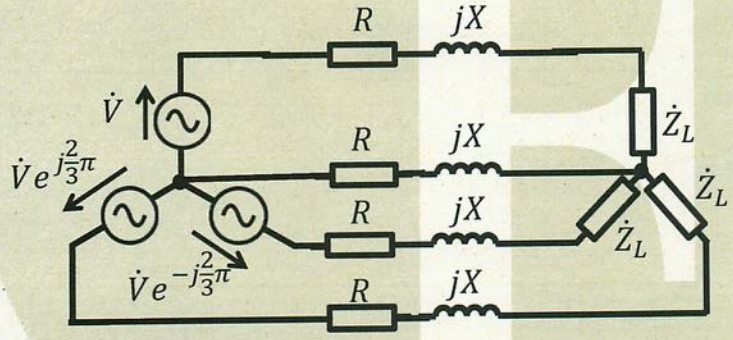
Fig. 1



(a) Single-phase two-wire



(b) Single-phase three-wire



(c) Three-phase four-wire

Fig. 2